

Suadeh 10303

SECTION 16.4

1. Match each parametrization with the corresponding surface in Figure 16.

- (a) $(u, \cos v, \sin v)$
- (b) $(u, u + v, v)$
- (c) (u, v^3, v)
- (d) $(\cos u \sin v, 3 \cos u \sin v, \cos v)$
- (e) $(u, u(2 + \cos v), u(2 + \sin v))$

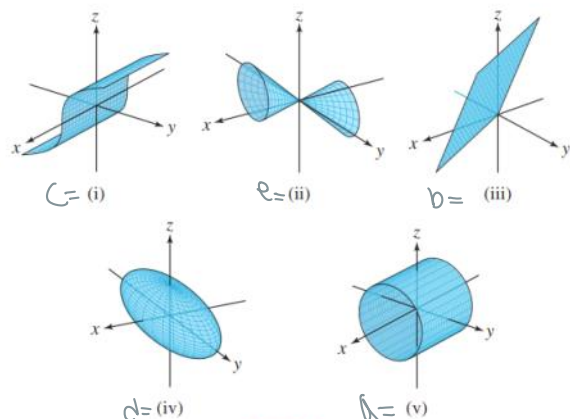


FIGURE 16

3. Show that $G(u, v) = (2u + 1, u - v, 3u + v)$ parametrizes the plane $2x - y - z = 2$. Then:

- (a) Calculate T_u , T_v , and $N(u, v)$.
- (b) Find the area of $S = G(D)$, where $D = \{(u, v) : 0 \leq u \leq 2, 0 \leq v \leq 1\}$.
- (c) Express $f(x, y, z) = yz$ in terms of u and v , and evaluate $\iint_S f(x, y, z) dS$.

$$f(x, y, z) = 2x - y - z = 2$$

$$x = 2u + 1$$

$$y = u - v$$

$$z = 3u + v$$

$$2(2u + 1) - (u - v) - (3u + v) = 2$$

$$4u + 2 - u + v - 3u - v = 2$$

$$\boxed{2 = 2} \text{ hence parametrizes the plane.}$$

a)

$$T_u = \frac{\partial G}{\partial u} = (2, 1, 3)$$

$$T_v = \frac{\partial G}{\partial v} = (-1, -1, 1)$$

$$u = \frac{v \cdot u}{\delta u} = (2, 1, 2)$$

$$T_v = \frac{\partial G}{\partial v} = (0, -1, 1)$$

$$N_{(uv)} = \begin{vmatrix} i & j & k \\ 2 & 1 & 3 \\ 0 & -1 & 1 \end{vmatrix} = i(4) - j(2) + k(-2)$$

$$N = (4, -2, -2)$$

b) for area $S = G(D)$ $|N| = 2\sqrt{6}$

$$A = \iint_D |N| \, du \, dv$$

Since $|N|$ is constant

$$A = 2\sqrt{6} \times \text{Area}(D)$$

$$0 \leq u \leq 2 \quad 0 \leq v \leq 1$$

$$A = 2\sqrt{6} \times 2 = \boxed{4\sqrt{6}} \text{ Am.}$$

c)

$$f(x, y, z) = yz \quad \text{in terms of } (u, v)$$

$$y = u - v, \quad z = 3u + v$$

$$f(x, y, z) = yz = (u - v)(3u + v) = 3u^2 - 2uv - v^2$$

- Surface integral:-

$$\iint_S f(x, y, z) \cdot dS = \iint_D f(u, v) \cdot |N| \cdot du \, dv$$

$$= 2\sqrt{6} \int_0^2 \int_0^1 (3u^2 - 2uv - v^2) \, dv \, du$$

$$= 2\sqrt{6} \int_0^2 \left(3u^2v - \frac{2uv^2}{2} - \frac{v^3}{3} \right) \Big|_0^1 \cdot du$$

$$= 2\sqrt{6} \int_0^2 \left(3u^2 - u - \frac{1}{3} \right) \cdot du$$

$$= 2\sqrt{6} \left(\frac{3u^3}{3} - \frac{u^2}{2} - \frac{u}{3} \right) \Big|_0^2 = 2\sqrt{6} \left[(2^3 - 0) - \frac{(2^2)}{2} - \frac{2}{3} \right]$$

$$= 2\sqrt{6} \left[8 - 2 - \frac{2}{3} \right]$$

$$= 2\sqrt{6} \left[\frac{24 - 6 - 2}{3} \right]$$

Ans. $\boxed{= \frac{32\sqrt{6}}{3}}$

5. Let $G(x, y) = (x, y, xy)$.

(a) Calculate $\mathbf{T}_x$, $\mathbf{T}_y$, and $\mathbf{N}(x, y)$.

(b) Let S be the part of the surface with parameter domain $\mathcal{D} = \{(x, y) : x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$. Verify the following formula and evaluate using polar coordinates:

$$\iint_S 1 \, dS = \iint_{\mathcal{D}} \sqrt{1 + x^2 + y^2} \, dx \, dy$$

(c) Verify the following formula and evaluate:

$$\iint_S z \, dS = \int_0^{\pi/2} \int_0^1 (\sin \theta \cos \theta) r^3 \sqrt{1 + r^2} \, dr \, d\theta$$

a) $G = (x, y, xy)$

$$\mathbf{T}_x = (1, 0, y)$$

$$\mathbf{T}_y = (0, 1, x)$$

$$\begin{aligned} \mathbf{N}(x, y) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & y \\ 0 & 1 & x \end{vmatrix} = \mathbf{i}(-y) - \mathbf{j}(x) + \mathbf{k}(1) \\ &= -y\mathbf{i} - x\mathbf{j} + \mathbf{k} \end{aligned}$$

b)

$$\iint_S 1 \cdot dS = \iint_{\mathcal{D}} \sqrt{1 + x^2 + y^2} \cdot dx \cdot dy$$

$$dS = |\mathbf{N}| \cdot dx \cdot dy = \sqrt{1 + x^2 + y^2} \cdot dx \cdot dy$$

$$\therefore x^2 + y^2 \leq 1, \quad x \geq 0, \quad y \geq 0$$

$$x = r \cos \theta \quad y = r \sin \theta$$

$$\therefore dx dy = r \cdot dr \cdot d\theta$$

$$\iint_S z \cdot dS = \int_0^{\pi/2} \int_0^1 \sqrt{1+r^2} \cdot r \cdot dr \cdot d\theta$$

Inner integral:-

$$I = \int_0^1 r \sqrt{1+r^2} \, dr$$

$$\text{let } u = 1+r^2 \quad \frac{du}{2} = r \cdot dr$$

$$I = \frac{1}{2} \int_1^2 \sqrt{u} \cdot du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} \Big|_1^2 = \frac{1}{3} (2^{3/2} - 1)$$

Now θ -Integration-

$$\int_0^{\pi/2} d\theta = \pi/2$$

$$\iint_S z \cdot dS = \frac{\pi}{2} \cdot \frac{1}{3} (2^{3/2} - 1) = \frac{\pi}{6} (2\sqrt{2} - 1)$$

$$c) \iint_S z \cdot dS = \int_0^{\pi/2} \int_0^1 (\sin \theta \cos \theta) r^3 \sqrt{1+r^2} \cdot dr \cdot d\theta$$

$$\text{since } z = xy = r^2 \cos \theta \sin \theta$$

$$z = r^2 \cos \theta \sin \theta$$

$$dS = \sqrt{1+r^2} \cdot r \cdot dr \cdot d\theta$$

$$z \, dS = (r^2 \sin \theta \cos \theta) \sqrt{1+r^2} \cdot r \cdot dr \cdot d\theta$$

Here verified

evaluate-

$$\iint_S z \cdot ds = \left(\int_0^{\pi/2} \sin \theta \cos \theta \cdot d\theta \right) \left(\int_0^1 \sqrt[3]{1+r^2} \cdot dr \right)$$

θ -integral

$$\int_0^{\pi/2} \sin \theta \cos \theta \cdot d\theta = \frac{1}{2} \int_0^{\pi/2} \sin 2\theta \cdot d\theta = \frac{1}{2}$$

after solving:-

$$\iint_S z \cdot ds = \frac{1}{2} \times \frac{2}{15} (\sqrt{2} + 1) = \frac{1}{15} (\sqrt{2} + 1)$$

In Exercises 7–10, calculate $\mathbf{T}_u$, $\mathbf{T}_v$, and $\mathbf{N}(u, v)$ for the parametrized surface at the given point. Then find the equation of the tangent plane to the surface at that point.

8. $\mathbf{G}(u, v) = (u^2 - v^2, u + v, u - v); \quad u = 2, \quad v = 3$

$$\mathbf{G}_u = \frac{\partial \mathbf{G}}{\partial u} = (2u, 1, 1)$$

$$\mathbf{G}_v = \frac{\partial \mathbf{G}}{\partial v} = (-2v, 1, -1)$$

$$\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2u & 1 & 1 \\ -2v & 1 & -1 \end{vmatrix} = (-2, -2, 10)$$

• point on surface $(u=2)(v=3)$

$$\mathbf{G}(2, 3) = (4 - 9, 5, -1) = (-5, 5, -1)$$

• equation of tangent line:-

$$\mathbf{N} \cdot [\langle \mathbf{r}, \mathbf{y}, \mathbf{z} \rangle - \mathbf{P}] = 0$$

$$\mathbf{N} = (-2, -2, 10) \text{ and } \mathbf{P} = (-5, 5, -1)$$

$$-2(x+5) - 2(y-5) + 10(z+1) = 0$$

$$-2x - 2y + 10z + 10 = 0$$

$$\boxed{x+y-5z=5}$$

10. $G(r, \theta) = (r \cos \theta, r \sin \theta, 1 - r^2); \quad r = \frac{1}{2}, \quad \theta = \frac{\pi}{4}$

$$G_r = (\cos \theta, \sin \theta, -2r)$$

$$G_\theta = (-r \sin \theta, r \cos \theta, 0)$$

$$N = G_r \times G_\theta = \begin{vmatrix} i & j & k \\ \cos \theta & \sin \theta & -2r \\ -r \sin \theta & r \cos \theta & 0 \end{vmatrix}$$

$$N = (2r^2 \cos \theta, 2r^2 \sin \theta, r)$$

$$r = 1/2 \quad \theta = \pi/4$$

$$\cos \pi/4 = \sin \pi/4 = \frac{\sqrt{2}}{2}$$

$$N = \left(\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, 1/2 \right)$$

- Point on surface-

$$\begin{aligned} G(1/2, \pi/4) &= \left(\frac{1}{2} \cos \pi/4, \frac{1}{2} \sin \pi/4, 1 - \frac{1}{2}^2 \right) \\ &= \left(\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, 3/4 \right) \end{aligned}$$

In Exercises 13–26, calculate $\iint_S f(x, y, z) dS$ for the given surface and function.

14. $G(r, \theta) = (r \cos \theta, r \sin \theta, \theta), \quad 0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi;$
 $f(x, y, z) = \sqrt{x^2 + y^2}$

$$G_r = (\cos \theta, \sin \theta, 0), \quad G_\theta = (-r \sin \theta, r \cos \theta, 1)$$

$$G_r \times G_\theta = (\sin \theta, -\cos \theta, r)$$

$$|G_r \times G_\theta| = \sqrt{\sin^2 \theta + \cos^2 \theta + r^2} = \sqrt{1 + r^2}$$

$$\iint_S f \cdot dS = \int_0^{2\pi} \int_0^1 (f(G(r, \theta))) |G_r \times G_\theta| \cdot dr \cdot d\theta$$

$$= \int_0^{2\pi} \int_0^1 r \sqrt{1 + r^2} \cdot dr \cdot d\theta$$

$$u = 1 + r^2 \quad du = 2r \, dr$$

$$\int_0^1 r \sqrt{1 + r^2} \cdot dr = \frac{1}{2} \int_1^2 u^{1/2} \, du = \frac{1}{3} (2\sqrt{2} - 1)$$

$$\theta \text{ integral} = 2\pi$$

$$\iint_S f \, dS = 2\pi \cdot \frac{1}{3} (2\sqrt{2} - 1) = \frac{2\pi}{3} (2\sqrt{2} - 1)$$

16. $y = 9 - z^2, \quad 0 \leq x \leq z \leq 3; \quad f(x, y, z) = 1$

$$y = g(x, z) = 9 - z^2 \quad g_x = 0, \quad g_z = -2z$$

$$\sqrt{1 + g_x^2 + g_z^2} = \sqrt{1 + 4z^2}$$

$$\text{So, } \iint_S f(x, y, z) \, dS = \iint_D \sqrt{1 + 4z^2} \cdot dx \cdot dz$$

$$= \int_0^3 \int_0^z \sqrt{1 + 4z^2} \cdot 1 \cdot 1 \cdot dx \cdot dz$$

1) 5

$$= \int_0^3 \int_0^2 \sqrt{1+4z^2} \cdot dz \cdot dz$$

$$= \int_0^3 \left[\sqrt{1+4z^2} \right]_0^2 dz$$

$$= \int_0^3 \sqrt{1+4z^2} \cdot z \cdot dz$$

let

$$u = 1+4z^2 \quad du = 8z \cdot dz$$

$$\int_0^3 \sqrt{1+4z^2} \cdot z \cdot dz = \int_1^{37} \frac{u^{1/2}}{8} \cdot du = 18.67$$

17. $x^2 + y^2 + z^2 = 1$, $x, y, z \geq 0$; $f(x, y, z) = x^2$.

$$\phi(\theta, \phi) = (\cos \theta \sin \phi, \sin \theta \sin \phi, \cos \phi)$$

$$0 \leq \theta \leq \pi/2 \quad 0 \leq \phi \leq \pi/2$$

$$\|n\| = \sin \phi$$

$$\iint_S x^2 \cdot ds = \int_0^{\pi/2} \int_0^{\pi/2} (\cos^2 \theta \sin^2 \phi) (\sin \phi) d\theta d\phi$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} \cos^2 \theta \sin^3 \phi d\phi d\theta$$

$$= \left(\int_0^{\pi/2} \cos^2 \theta \cdot d\theta \right) \left(\int_0^{\pi/2} \sin^3 \phi \cdot d\phi \right) = \left(\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right) \Big|_0^{\pi/2} \left(\frac{-\sin^2 \phi \cos \phi - 2/3 \cos \phi}{3} \right) \Big|_0^{\pi/2}$$

$$= \pi/4 \cdot 2/3 = \boxed{\pi/6}$$

21. Part of the plane $x + y + z = 1$, where $x, y, z \geq 0$;

$$f(x, y, z) = z$$

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \cdot dx \cdot dy$$

$$g_x = -1 \quad \text{and} \quad g_y = -1$$

$$\sqrt{1 + g_x^2 + g_y^2} = \sqrt{3}$$

$$f(x, y, g(x, y)) = 1 - x - y$$

$$\iint_S f(x, y, z) \cdot dS = \int_0^1 \int_0^{1-y} (1 - x - y) \sqrt{3} \cdot dx \cdot dy$$

$$= \sqrt{3} \int_0^1 \left(x - \frac{x^2}{2} - yx \right) \Big|_{x=0}^{1-y} dy$$

$$= \sqrt{3} \int_0^1 \left((1-y)^2 - \frac{(1-y)^2}{2} \right) dy = \frac{\sqrt{3}}{2} \int_0^1 (1 - 2y + y^2) \cdot dy$$

$$= \frac{\sqrt{3}}{2} \left(y - y^2 + \frac{y^3}{3} \right) \Big|_0^1 = \sqrt{3}/6$$

22. Part of the plane $x + y + z = 0$ contained in the cylinder $x^2 + y^2 = 1$;

$$f(x, y, z) = z^2$$

$$z = -x - y \quad z = g(x, y)$$

$$g_x = -1 \quad g_y = -1$$

$$\|n\| = \sqrt{1 + g_x^2 + g_y^2} = \sqrt{3}$$

$$\begin{aligned}
 \iint_S z^2 \, dS &= \iint_D (-x-y)^2 \sqrt{3} \, dx \, dy \\
 &= \int_0^{2\pi} \int_0^1 (-r \cos \theta - r \sin \theta)^2 \sqrt{3} \, r \, dr \, d\theta \\
 &= \int_0^{2\pi} (1 + \sin 2\theta) \, d\theta \left(\sqrt{3} \int_0^1 r^3 \, dr \right) \\
 &= \left(\theta - \frac{\cos 2\theta}{2} \Big|_0^{2\pi} \right) \left(\sqrt{3} \frac{r^4}{4} \Big|_0^1 \right) = \boxed{\frac{\pi \sqrt{3}}{2}} \text{ Ans.}
 \end{aligned}$$

26. Part of the unit sphere centered at the origin, where $x \geq 0$ and $|y| \leq x$; $f(x, y, z) = x$

$$\phi(\theta, \phi) = (\cos \theta \sin \phi, \sin \theta \sin \phi, \cos \phi)$$

$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4} \quad 0 \leq \phi \leq \pi$$

$$|n| = r^2 \sin \phi = \sin \phi$$

$$f(\phi(\theta, \phi)) = x = \cos \theta \sin \phi$$

$$\iint_S f(x, y, z) \, dS = \iint_D f(\phi(\theta, \phi)) |n| \, d\theta \, d\phi$$

$$= \int_0^\pi \int_{-\pi/4}^{\pi/4} \cos \theta \sin \phi \, d\theta \, d\phi$$

$$= \left(\frac{\theta}{2} - \frac{\sin 2\theta}{4} \Big|_{\theta=-\pi/4}^{\pi/4} \right) \left(\sin \phi \Big|_{\phi=0}^{\pi/4} \right) = \frac{\pi}{2} \frac{\sqrt{2}}{2} = \frac{\pi}{\sqrt{2}}$$

27. A surface S has a parametrization $G(u, v)$ with domain $0 \leq u \leq 2, 0 \leq v \leq 4$ such that the following partial derivatives are constant:

$$\frac{\partial G}{\partial u} = \langle 2, 0, 1 \rangle, \quad \frac{\partial G}{\partial v} = \langle 4, 0, 3 \rangle$$

What is the surface area of S ?

$$n = T_u \times T_v = \begin{vmatrix} i & j & k \\ 2 & 0 & 1 \\ 4 & 0 & 3 \end{vmatrix} = \langle 0, -2, 0 \rangle \Rightarrow |n| = 2$$

$$\text{Area}(S) = \iint_D |n| \, du \, dv = \iint_D 2 \, du \, dv = 2 \cdot \text{Area}(D) = 2 \cdot 2 \cdot 4 = 16$$

29. Calculate $\iint_S (xy + e^z) \, dS$, where S is the triangle in Figure 18 with vertices $(0, 0, 3)$, $(1, 0, 2)$, and $(0, 4, 1)$.

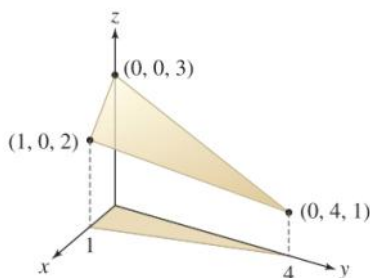


FIGURE 18

$$A = (0, 0, 3), \quad B = (0, 4, 1), \quad C = (1, 0, 2)$$

$$AB \times AC = \langle 0, 4, -2 \rangle \times \langle 1, 0, -1 \rangle$$

$$= \begin{vmatrix} i & j & k \\ 0 & 4 & -2 \\ 1 & 0 & -1 \end{vmatrix} = -4i - 2j - 4k = -2\langle 2, 1, 2 \rangle$$

Plane passing through $A = (0, 0, 3)$ and perpendicular to vector $\langle 2, 1, 2 \rangle$

$$\langle x - 0, y - 0, z - 3 \rangle \cdot \langle 2, 1, 2 \rangle = 0$$

$$2x + y + 2(z - 3) = 0$$

$$2x + y + 2z = 6$$

$$z = g(x, y) = -x - \frac{1}{2}y + 3$$

Now,

$$\iint_S f(x, y, z) \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy$$

$$g_x = -1, \quad g_y = -\frac{1}{2} \Rightarrow \sqrt{1 + g_x^2 + g_y^2}$$

$$= \sqrt{1 + (-1)^2 + \left(-\frac{1}{2}\right)^2} = \frac{3}{2}$$

$$f(x, y, g(x, y)) = xy + e^z = xy + e^{-x - y/2 + 3}$$

$$\iint_S f(x, y, z) \, dS = \int_0^1 \int_0^{-4x+4} (xy + e^{-x-y/5}) \frac{3}{2} \, dy \, dx$$

$$= \int_0^1 \left. \frac{3xy^2}{4} - 3e^{-x-y/5} \right|_0^{-4x+4} \, dx$$

$$= \int_0^1 (12x^3 - 24x^2 + 12x - 3e^{x+1} + 3e^{-x+3}) \, dx$$

$$= 3x^4 - 8x^3 + 6x^2 - 3e^{x+1} - 3e^{-x+3} \Big|_0^1$$

$$= (1 - 3e^2 - 3e^2) - (-3e - 3e^3) = \boxed{25.611}$$

35. What is the area of the portion of the plane $2x + 3y + 4z = 28$ lying above the domain D in the xy -plane in Figure 19 if $\text{Area}(D) = 5$?

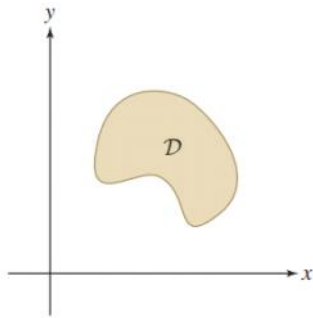


FIGURE 19

$$z = g(x, y) = -\frac{2x}{4} - \frac{3y}{4} + 7$$

$$\sqrt{1 + g_x^2 + g_y^2} = \sqrt{1 + \left(-\frac{1}{2}\right)^2 + \left(-\frac{3}{4}\right)^2} = \frac{\sqrt{29}}{4}$$

$$\begin{aligned} \text{Area}(S) &= \iint_S 1 \, dS = \iint_D \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy \\ &= \iint_D \frac{\sqrt{29}}{4} \, dx \, dy = \frac{\sqrt{29}}{4} \text{Area}(D) \\ &= \frac{\sqrt{29}}{4} \cdot 5 = \boxed{6.7} \end{aligned}$$

41. Prove a famous result of Archimedes: The surface area of the portion of the sphere of radius R between two horizontal planes $z = a$ and $z = b$ is equal to the surface area of the corresponding portion of the circumscribed cylinder (Figure 22).

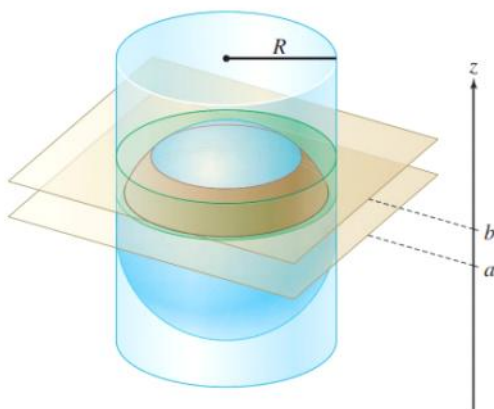


FIGURE 22

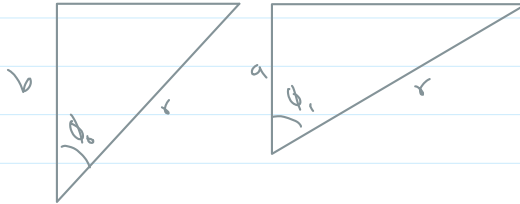
$$\phi(\theta, \phi) = (r \cos \theta \sin \phi, r \sin \theta \sin \phi, r \cos \phi)$$

$$0 \leq a < b$$

$$D_1: 0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \phi_1, \quad \text{angle } \phi_1 \text{ and } \phi_0 \text{ are determined by:}$$

$$\cos \phi_0 = \frac{b}{r} \quad \phi_0 = \cos^{-1}(b/r)$$

$$\phi_1 = \cos^{-1}(a/r)$$



$$\text{Area}(S_1) = \iint_{D_1} |n| \, d\phi \, d\theta = \int_0^{2\pi} \int_{\phi_0}^{\phi_1} r^2 \sin \phi \, d\phi \, d\theta$$

$$= \left(\int_0^{2\pi} r^2 \, d\theta \right) \left(\int_{\phi_0}^{\phi_1} \sin \phi \, d\phi \right)$$

$$= 2\pi r^2 \left(-\cos \phi \right) \Big|_{\phi = \cos^{-1}(b/r)}^{\phi = \cos^{-1}(a/r)} = 2\pi r^2 \left(-\frac{a}{r} + \frac{b}{r} \right)$$

$$= 2\pi r (b-a)$$

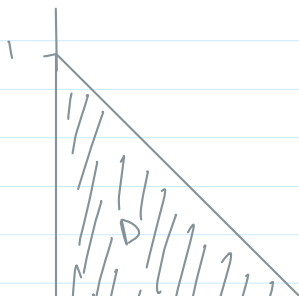
$$\text{Area}(S_2) = 2\pi r (b-a)$$

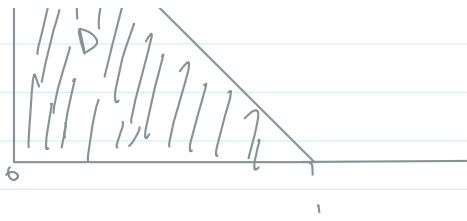
$$\text{Area}(S_1) = \text{Area}(S_2)$$

SECTION 16.5

11. $\mathbf{F} = y^2 \mathbf{i} + 2y \mathbf{j} - x \mathbf{k}$, portion of the plane $x + y + z = 1$ in the octant $x, y, z \geq 0$, upward-pointing normal

$$\phi(x, y) = (x, y, 1-x-y)$$





$$T_x = \frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x} (x, y, 1-x-y) = \langle 1, 0, -1 \rangle$$

$$T_y = \langle 0, 1, -1 \rangle$$

$$n = \begin{vmatrix} i & j & k \\ \phi & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = i + j + k = \langle 1, 1, 1 \rangle$$

$$F \cdot n = \langle y^2, 2, -x \rangle \cdot \langle 1, 1, 1 \rangle = y^2 + 2 - x$$

$$\iint_S F \cdot ds = \iint_D F(\phi(x, y)) \cdot n \, dx \, dy$$

$$= \int_0^1 \int_0^{1-y} (y^2 + 2 - x) \, dx \, dy = \int_0^1 \left. y^2 x + 2x - \frac{x^2}{2} \right|_0^{1-y} dy$$

$$= \int_0^1 \left(y^2 - y^3 + 2(1-y) - \frac{(y-1)^2}{2} \right) \cdot dy$$

$$= \left. \frac{y^3}{3} - \frac{y^4}{4} - (1-y)^2 - \frac{(y-1)^3}{6} \right|_0^1 = \left(\frac{1}{3} - \frac{1}{4} \right) + \left(1 - \frac{1}{6} \right) = \frac{11}{12}$$

13. $\mathbf{F} = \langle xz, yz, z^{-1} \rangle$, disk of radius 3 at height 4 parallel to the xy -plane, upward-pointing normal

$$\mathbf{Q}(\theta, r) = (r \cos \theta, r \sin \theta, 4)$$

$$T_{\mathbf{Q}} = \langle -r \sin \theta, r \cos \theta, 0 \rangle$$

$$T_r = \langle \cos \theta, \sin \theta, 0 \rangle$$

$$n = T_\theta \times T_r = \begin{vmatrix} i & j & k \\ -r \sin \theta & r \cos \theta & 0 \\ \cos \theta & \sin \theta & 0 \end{vmatrix} = (-r \sin^2 \theta - r \cos^2 \theta) k$$

$$= -rk = \langle 0, 0, -r \rangle$$

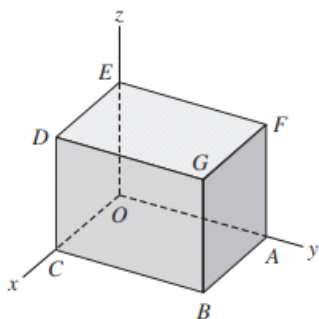
$$n = \langle 0, 0, r \rangle$$

$$F(\phi(\theta, r)) = \langle xz, yz, z^{-1} \rangle = \langle r \cos \theta \cdot \frac{1}{r}, r \sin \theta \cdot \frac{1}{r}, \frac{1}{r} \rangle$$

$$F \cdot n = \langle r \cos \theta, r \sin \theta, \frac{1}{r} \rangle \cdot \langle 0, 0, r \rangle = \frac{1}{r}$$

$$\iint_S F \cdot ds = \int_0^{2\pi} \int_0^3 \frac{r}{r} \cdot dr = 2\pi \left. \frac{r^2}{2} \right|_0^3 = \frac{9\pi}{1}$$

15. $F = \langle 0, 0, e^{y+z} \rangle$, boundary of unit cube $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$, outward-pointing normal



$S_1 = \text{Face } OABC$

$S_2 = DGEF$

$S_3 = ABGF$

$S_4 = OCDE$

$S_5 = BCDG$

$S_6 = OAFE$

$S_1:-$

$$\Phi_1(x, y) = (x, y, 0)$$

$$\text{w.s. } n_1 = \langle 0, 0, -1 \rangle$$

$$F_1(\Phi_1(x, y)) \cdot n_1 = \langle 0, 0, e^y \rangle \cdot \langle 0, 0, -1 \rangle = -e^y$$

$$F_1(\phi_1(x, y)) \cdot n_1 = \langle 0, 0, e^y \rangle \cdot \langle 0, 0, -1 \rangle = -e^y$$

$S_2:-$

$$\phi_2(x, y) = (x, y, 1)$$

$$F \cdot n = \langle 0, 0, e^{y+1} \rangle \cdot \langle 0, 0, 1 \rangle = e^{y+1}$$

on all other- S_3 to S_6

$$F(\phi_1(x, y)) \cdot n = 0$$

Now,

$$\iint_S F \cdot ds = \iint_{S_1} F \cdot ds + \iint_{S_2} F \cdot ds$$

$$= \int_0^1 \int_0^1 -e^y \cdot dx \cdot dy + \int_0^1 \int_0^1 e^{y+1} \cdot dx \cdot dy$$

$$= \int_0^1 e^y(e-1) \cdot dy = \boxed{(e-1)^2}$$

17. $F = \langle y, z, 0 \rangle$, $G(u, v) = (u^3 - v, u + v, v^2)$, $0 \leq u \leq 2$,
 $0 \leq v \leq 3$, downward-pointing normal

$$T_u = \frac{\partial G}{\partial u} = \frac{\partial}{\partial u} (u^3 - v, u + v, v^2) = \langle 3u^2, 1, 0 \rangle$$

$$T_v = \frac{\partial G}{\partial v} = \frac{\partial}{\partial v} (u^3 - v, u + v, v^2) = \langle -1, 1, 2v \rangle$$

$$T_u \times T_v = \begin{vmatrix} i & j & k \\ 3u^2 & 1 & 0 \\ -1 & 1 & 2v \end{vmatrix} = 2v \hat{i} - (6u^2 v) \hat{j} + (3u^2 + 1) \hat{k}$$

$$T_u \times T_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3u^2 & 1 & 0 \\ -1 & 1 & 2v \end{vmatrix} = 2v\mathbf{i} - (6u^2v)\mathbf{j} + (3u^2+1)\mathbf{k}$$

$$= \langle 2v, -6u^2v, 3u^2+1 \rangle$$

since, in physics convention, z-coordinate is -ve.

$$\mathbf{n} = \langle -2v, 6u^2v, -3u^2-1 \rangle$$

Now

$$\mathbf{F} \cdot \mathbf{n} = \langle y, z, 0 \rangle \cdot \langle u+v, v^2, 0 \rangle$$

$$\mathbf{F}(\phi(u, v)) \cdot \mathbf{n}(u, v) = \langle u+v, v^2, 0 \rangle \cdot \langle -2v, 6u^2v, -3u^2-1 \rangle$$

$$= -2v(u+v) + 6u^2v \cdot v^2 + 0$$

$$= -2v^2 - 2vu + 6u^2v^3 \text{ Am.}$$

$$\iint_S \mathbf{F} \cdot d\mathbf{s} = \iint_D \mathbf{F}(\phi(u, v)) \cdot \mathbf{n}(u, v) du \cdot dv = \int_0^3 \int_0^2 (-2uv - 2v^2 + 6u^2v^3) du \cdot dv$$

$$= \int_0^3 \left. -u^2v - 2v^2u + 2u^3v^3 \right|_{u=0}^2 dv = \int_0^3 (16v^3 - 4v^2 - 4v) dv$$

$$\boxed{= 270}$$

19. Let $\mathbf{e}_r = \langle x/r, y/r, z/r \rangle$ be the unit radial vector, where $r = \sqrt{x^2 + y^2 + z^2}$. Calculate the integral of $\mathbf{F} = e^{-r} \mathbf{e}_r$ over:

(a) the upper hemisphere of $x^2 + y^2 + z^2 = 9$, outward-pointing normal.

(b) the octant $x \geq 0, y \geq 0, z \geq 0$ of the unit sphere centered at the origin.

$$\phi - x = 3 \cos \theta \sin \phi$$

$$y = 3 \sin \theta \sin \phi$$

$$y = 3 \sin \psi \sin \phi$$

$$z = 3 \cos \phi$$

$$n = 9 \sin \phi e_r \quad \therefore r = 3$$

$$F \cdot n = e^{-r} e_r \cdot n = e^{-3} e_r \cdot 9 \sin \phi e_r = 9 e^{-3} \sin \phi e_r = 9 e^{-3} \sin \phi$$

$$\begin{aligned} \iint_S F \cdot ds &= \iint_D (F \cdot n) \cdot d\phi dQ = \int_0^{2\pi} \int_0^{\pi/2} 9 e^{-3} \sin \phi \cdot d\phi dQ \\ &= 18\pi e^{-3} \int_0^{\pi/2} \sin \phi d\phi = 18\pi e^{-3} \left(-\cos \phi \Big|_0^{\pi/2} \right) = 18\pi e^{-3} \end{aligned}$$

b) Unit octant of sphere :-

$$\begin{aligned} \phi : \quad x &= \cos Q \sin \phi \\ y &= \sin Q \sin \phi \\ z &= \cos \phi \end{aligned}$$

Outward pointing normal:-

$$n = \sin \phi e_r$$

$$F \cdot n = e^{-r} \cdot e_r \cdot n = e^{-1} \sin \phi e_r = e^{-1} \sin \phi$$

$$\iint_S F \cdot ds = \iint_D (F \cdot n) d\phi dQ = \int_0^{\pi/2} \int_0^{\pi/2} e^{-1} \sin \phi \cdot d\phi dQ$$

$$= \frac{\pi}{2} e^{-1} \int_0^{\pi/2} \sin \phi d\phi = \frac{\pi}{2} e^{-1} \left(-\cos \phi \Big|_0^{\pi/2} \right) = \frac{\pi}{2} e^{-1}$$

21. The electric field due to a point charge located at the origin in $\mathbf{R}^3$ is $\mathbf{E} = k \frac{\mathbf{e}_r}{r^2}$, where $r = \sqrt{x^2 + y^2 + z^2}$ and k is a constant. Calculate the flux of $\mathbf{E}$ through the disk D of radius 2 parallel to the xy -plane with center $(0, 0, 3)$.

$$\phi(\hat{r}, \theta) = (r \cos \theta, r \sin \theta, z)$$

$$T_r = \frac{\partial \phi}{\partial r} = \langle \cos \theta, \sin \theta, 0 \rangle$$

$$T_\theta = \frac{\partial \phi}{\partial \theta} = \langle -r \sin \theta, r \cos \theta, 0 \rangle$$

$$n = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos \theta & \sin \theta & 0 \\ -r \sin \theta & r \cos \theta & 0 \end{vmatrix} = \langle 0, 0, r \rangle$$

$$\begin{aligned} \mathbf{E} \cdot n &= 3k \frac{\hat{r}}{r^3} = \frac{k r}{r^3} \langle x, y, z \rangle \cdot \langle 0, 0, 0 \rangle \\ &= \frac{z k \hat{r}}{r^3} \end{aligned}$$

$$\mathbf{E} n = 3k \frac{\hat{r}}{r^3} \quad \text{and} \quad r = \sqrt{\hat{r}^2 + 9}$$

$$\iint_D \mathbf{E} \cdot d\mathbf{s} = \int_0^{2\pi} \int_0^2 \frac{3kr}{(\hat{r}^2 + 9)^{3/2}} \cdot dr \cdot d\theta = 6\pi k \int_0^2 \frac{\hat{r}}{(\hat{r}^2 + 9)^{3/2}} d\hat{r}$$

$$u = \hat{r}^2 + 9 \quad \text{and} \quad \frac{1}{2} du = \hat{r} \cdot d\hat{r}$$

$$\begin{aligned} \iint_D \mathbf{E} \cdot d\mathbf{s} &= 3\pi k \int_9^{13} \frac{du}{u^{3/2}} = -6\pi k u^{-1/2} \Big|_9^{13} \\ &= \left(2 - \frac{6}{\sqrt{13}} \right) \pi k \end{aligned}$$

In Exercises 25–28, a net is dipped in a river. Determine the flow rate of water across the net if the velocity vector field for the river is given by $\mathbf{v}$ and the net is described by the given equations.

25. $\mathbf{v} = \langle x - y, z + y + 4, z^2 \rangle$, net given by $x^2 + z^2 \leq 1$, $y = 0$, oriented in the positive y -direction

$$\text{flow} = \iint_S \mathbf{v} \cdot d\mathbf{S} = \iint_S \mathbf{v} \cdot \mathbf{n} \cdot dS$$

on $y=0$:-

$$\mathbf{v} = \langle x, 0, z \rangle = \langle x - 0, z + 0 + 4, z^2 \rangle = \langle x, z + 4, z^2 \rangle$$

$$\mathbf{v} \cdot \mathbf{n} = \langle x, z + 4, z^2 \rangle \cdot \langle 0, 1, 0 \rangle = z + 4$$

$$\text{flow} = \iint_S (z + 4) \cdot dA$$

$$= \int_0^{2\pi} \int_0^1 (r \sin \theta + 4) \cdot r \cdot dr \cdot d\theta$$

$$= \int_0^{2\pi} \int_0^1 (r^2 \sin \theta + 4r) \cdot dr \cdot d\theta$$

$$= \int_0^{2\pi} \left(\frac{\sin \theta}{3} + 2 \right) \cdot d\theta = -\frac{1}{3} + 4\pi + \frac{1}{3} = \boxed{4\pi} \text{ Am.}$$

31. Prove that if S is the part of a graph $z = g(x, y)$ lying over a domain D in the xy -plane, then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left(-F_1 \frac{\partial g}{\partial x} - F_2 \frac{\partial g}{\partial y} + F_3 \right) dx dy$$

$$\phi(x, y) = (x, y, g(x, y)), (x, y) \in D$$

$$T_x = \frac{\partial \phi}{\partial x} = \left\langle 1, 0, \frac{\partial g}{\partial x} \right\rangle$$

$$T_y = \frac{\partial \phi}{\partial y} = \left\langle 0, 1, \frac{\partial g}{\partial y} \right\rangle$$

$$n = \begin{vmatrix} i & j & k \\ 1 & 0 & \frac{\partial g}{\partial x} \\ 0 & 1 & \frac{\partial g}{\partial y} \end{vmatrix} = -\frac{\partial g}{\partial x} i - \frac{\partial g}{\partial y} j + k$$

$$F \cdot n = \langle F_1, F_2, F_3 \rangle \cdot \left\langle -\frac{\partial g}{\partial x}, -\frac{\partial g}{\partial y}, 1 \right\rangle$$

$$= -F_1 \frac{\partial g}{\partial x} - F_2 \frac{\partial g}{\partial y} + F_3$$

$$\iint_S F \cdot ds = \iint_D (F \cdot n) \cdot dx \cdot dy = \iint_D \left(-F_1 \frac{\partial g}{\partial x} - F_2 \frac{\partial g}{\partial y} + F_3 \right) \cdot dx \cdot dy$$

34. Find the rate of heat flow out of a sphere of radius 1 meter inside a large cube of copper ($K = 400$ kilowatts/m-k) with temperature function given by $w(x, y, z) = 20 - 5(x^2 + y^2 + z^2)^\circ\text{C}$.

$$\text{heat flow} = -k \nabla w$$

$$\nabla w = \left\langle \frac{\partial w}{\partial x}, \frac{\partial w}{\partial y}, \frac{\partial w}{\partial z} \right\rangle$$

$$\nabla w = \langle -10x, -10y, -10z \rangle$$

$$F = -k \nabla w = -400 \langle -10x, -10y, -10z \rangle = \langle 4000x, 4000y, 4000z \rangle$$

heat flow out of sphere =

$$\iint_S F \cdot ds$$

Divergence theorem-

Divergence theorem:-

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} 4000x + \frac{\partial}{\partial y} 4000y + \frac{\partial}{\partial z} 4000z = 12000$$

$$\iiint_V 12000 dV = 12000 \cdot V$$

$$\therefore V = \frac{4\pi r^3}{3} = \frac{4\pi}{3}$$

So, rate of heat flow = 16000π kilowatt.

36. A point mass m is located at the origin. Let Q be the flux of the gravitational field $\mathbf{F} = -Gm \frac{\mathbf{e}_r}{r^2}$ through the cylinder $x^2 + y^2 = R^2$ for $a \leq z \leq b$, including the top and bottom (Figure 18). Show that $Q = -4\pi Gm$ if $a < 0 < b$ (m lies inside the cylinder) and $Q = 0$ if $0 < a < b$ (m lies outside the cylinder).

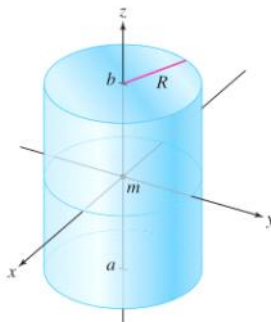


FIGURE 18

$S_1 \rightarrow$ cylinder

$S_2 \rightarrow$ Top

$S_3 \rightarrow$ bottom

$$S_1 \quad \phi_1(\theta, z) = (R \cos \theta, R \sin \theta, z) \\ 0 \leq \theta \leq 2\pi \quad a \leq z \leq b \\ \mathbf{n} = R \langle \cos \theta, \sin \theta, 0 \rangle$$

$S_2:-$

$$\phi(\theta, r) = (r \cos \theta, r \sin \theta, b) = 0 \leq \theta \leq 2\pi$$

2:-

$$\phi(Q, r) = (r \cos Q, r \sin Q, b) = 0 \leq Q \leq 2\pi$$

$$n = \langle 0, 0, r \rangle$$

S₃:-

$$\phi_3(Q, r) = (r \cos Q, r \sin Q, a), \quad 0 \leq Q \leq 2\pi$$

$$0 \leq r \leq R$$

$$n = \langle 0, 0, -r \rangle$$

$$Q = \iint_S F \cdot dS = \iint_{S_1} F \cdot dS + \iint_{S_2} F \cdot dS + \iint_{S_3} F \cdot dS$$

$$F(\phi_1(Q, z)) = -Gm \frac{e_v}{r^2} = -\frac{Gm}{(R^2 + z^2)^{3/2}} \langle R \cos Q, R \sin Q, z \rangle$$

$$F(\phi_1(Q, z)) \cdot n(Q, z) = -\frac{Gm}{(R^2 + z^2)^{3/2}} \langle R \cos Q, R \sin Q, z \rangle \cdot R \langle \cos Q, \sin Q, 0 \rangle = -\frac{Gm R^2}{(R^2 + z^2)^{3/2}}$$

$$\iint_{S_1} F \cdot dS = \int_0^{2\pi} \int_a^b -\frac{Gm R^2}{(R^2 + z^2)^{3/2}} dz \cdot dQ = -2\pi Gm R^2 \int_a^b \frac{dz}{(R^2 + z^2)^{3/2}}$$

we substitute $z = R \tan t$.

$$\iint_{S_1} F \cdot dS = -2\pi Gm R^2 \int_{\tan^{-1} a/R}^{\tan^{-1} b/R} \frac{\cos t}{R^2} dt = 2\pi Gm \sin t \Big|_{\tan^{-1} a/R}^{\tan^{-1} b/R}$$

$$= -2\pi Gm \left(\frac{b}{\sqrt{b^2 + R^2}} - \frac{a}{\sqrt{a^2 + R^2}} \right)$$

S₂:-

$$F(\phi_2(Q, r)) = -Gm \frac{e_v}{r^2} = -\frac{Gm}{(r^2 + b^2)^{3/2}} \langle r \cos Q, r \sin Q, b \rangle$$

Here

$$= -\frac{Gm}{(r^2 + b^2)^{3/2}} \cdot \langle r \cos Q, r \sin Q, b \rangle \cdot \langle 0, 0, r \rangle$$

$$= -\frac{Gm b r}{(r^2 + b^2)^{3/2}}$$

$$= -\frac{Cmbv}{(r^2+b^2)^{3/2}}$$

$$\begin{aligned}\iint_{S_2} \mathbf{F} \cdot d\mathbf{s} &= -\pi Cmb \int_{b^2}^{R^2+b^2} \frac{dt}{t^{3/2}} = 2\pi Cmb \frac{1}{\sqrt{t}} \bigg|_{t=b^2}^{R^2+b^2} \\ &= 2\pi Cmb \left(\frac{1}{\sqrt{b^2+R^2}} - \frac{1}{b} \right)\end{aligned}$$

$S_3:-$

$$\mathbf{F}(\phi_3(\theta, r)) = -Cm \frac{er}{r^2} = -\frac{Cm}{(r^2+a^2)^{3/2}} \langle r \cos \theta, r \sin \theta, a \rangle$$

$$\mathbf{F}(\phi_3(\theta, r)) \cdot \mathbf{r}(\theta, r) = -\frac{Cm}{(r^2+a^2)^{3/2}} \cdot \langle r \cos \theta, r \sin \theta, a \rangle \cdot \langle 0, 0, -r \rangle = \frac{Cmar}{(r^2+a^2)^{3/2}}$$

by the same computation of S_2 we get ($a < 0$).

$$(\sqrt{a^2} = -a)$$

$$\begin{aligned}\iint_{S_3} \mathbf{F} \cdot d\mathbf{s} &= \int_0^{2\pi} \int_b^R \frac{Cmar}{(r^2+a^2)^{3/2}} dr \cdot d\theta = -2\pi Cma \frac{1}{\sqrt{t}} \bigg|_{t=a^2}^{R^2+a^2} \\ &= -2\pi Cma \left(\frac{1}{\sqrt{R^2+a^2}} - \frac{1}{\sqrt{a^2}} \right) = -2\pi Cma \left(\frac{1}{\sqrt{R^2+a^2}} + \frac{1}{a} \right)\end{aligned}$$

Now,

$$\begin{aligned}Q &= -2\pi Cm \left(\frac{b}{\sqrt{b^2+R^2}} - \frac{a}{\sqrt{a^2+R^2}} \right) + 2\pi Cmb \left(\frac{1}{\sqrt{b^2+R^2}} - \frac{1}{b} \right) - 2\pi Cma \left(\frac{1}{\sqrt{R^2+a^2}} + \frac{1}{a} \right) \\ &= -2\pi Cm - 2\pi Cm = -4\pi Cm\end{aligned}$$

if $0 < a < b$

$$\iint_{S_3} \mathbf{F} \cdot d\mathbf{s} = -2\pi Cma \left(\frac{1}{\sqrt{R^2+a^2}} - \frac{1}{\sqrt{a^2}} \right) = -2\pi Cma \left(\frac{1}{\sqrt{R^2+a^2}} - \frac{1}{a} \right)$$

Therefore, -

$$Q = -2\pi \epsilon_0 \left(\frac{b}{\sqrt{b^2 + R^2}} - \frac{a}{\sqrt{a^2 + R^2}} \right) + 2\pi \epsilon_0 \frac{ab}{R} \left(\frac{1}{\sqrt{b^2 + R^2}} - \frac{1}{b} \right) - 2\pi \epsilon_0 \frac{a}{R} \left(\frac{1}{\sqrt{R^2 + a^2}} - \frac{1}{a} \right)$$
$$= -2\pi \epsilon_0 + 2\pi \epsilon_0 = \boxed{0}$$